

Image Prompt Surrounded by stacks of papers and old books in a sparsely furnished apartment room in St. Petersburg, a heavily bearded Grigori Perelman is deeply engrossed in scribbling complex mathematical equations on a notepad.

Text Search

Model Responses I'll help you gather information and reference images for creating a detailed image description about Grigoriy Yakovlevich Perel'man. Let me start by searching for background information

Tool Use `<tool_call>{"name": "text_search", "arguments": {"q": "Grigory Yegorovich Perlman, mathematician, biography", "hl": "en", "top_k": 2}}</tool_call>`

Tool Output `<observation>` [1] Grigory Perelman - Wikipedia
Grigory Yegorovich Perelman is a Russian mathematician, born on June 13, 1966, in Leningrad, former Soviet Union...
[2] The enigmatic mathematical genius Perelman: Say what you want | CAM-Net Digest...
`</observation>`



Image Search

Model Responses Now I have a good understanding of Grigoriy Perel'man - he's a Russian mathematician born in 1966 in Leningrad (st. Petersburg), known for his work on the Poincaré conjecture and other Millennium Prize problems. He's described as a mysterious mathematician who has turned down major awards. Let me search for reference images.

Tool Use `<tool_call>{"name": "search_image", "arguments": {"q": "Grigory Perelman, mathematician, Russian, Brazos", "hl": "en", "num": 10}}</tool_call>`

Tool Output `<observation>` [1] Grigory Perelman - Wikipedia, the free encyclopedia
[2] A Russian genius who loves numbers more than money - Science & Technology Window - RFI - Radio France Internationale
[3] [The genius in beggar's clothes] The Russian mathematician who refused a million-dollar prize but remained anonymous...
[4] Russian mathematician Perelman refuses a \$1 million prize - Mysterious Earth Science | Nature...
[5] Grigory Perelman, the mathematician who refused the Fields Medal and the \$1 million Clay Prize.
.....
`</observation>`



Image Judgement

Reference Images



Refined Recaption

Thinking `<think>` **Analyzing image_1:** This shows Grigoriy Perel'man with his distinctive appearance - he has long, dark brown wavy hair that flows past his shoulders...

Analyzing image_2: This image shows Perel'man with an older appearance - his hair has turned predominantly gray/white and is wild and curly, flowing outward in an untamed manner...

Integration strategy: I need to combine these visual characteristics with the scene description of a sparsely furnished apartment in St. Petersburg, surrounded by papers and old books, writing mathematical equations. The scene should capture his distinctive appearance - the long flowing hair (graying as in image_2), the full beard (also graying), deep-set eyes, oval face, and scholarly yet bohemian appearance. The setting reflects his reclusive, dedicated mathematical lifestyle.`</think>`

Recaption `<recaption>`In a sparsely furnished apartment room in St. Petersburg, illuminated by soft natural light filtering through a single window, sits the legendary mathematician Grigoriy Perel'man, deeply absorbed in his work. Preserve the exact facial features of Grigoriy Perel'man as shown in image_1 and image_2: an oval-shaped face with prominent cheekbones, deep-set eyes with a characteristic slight squint, thick dark eyebrows now peppered with gray, and a full, flowing beard that has transitioned from reddish-brown to gray-brown, wild and untamed like the hair that flows past his shoulders in image_2. His complexion is fair with a weathered, sun-touched quality, and his expression carries the intense, contemplative focus of a brilliant mind at work. Maintain his distinctive bohemian scholarly appearance as referenced in both images...`</recaption>`

Ground Truth



Generated Images

